

Complete EDA Guide for Students

Step-by-Step Exploratory Data Analysis (EDA) Guide with Explanations, Code & Visualizations

What is EDA?

EDA (Exploratory Data Analysis) is the process of understanding, cleaning, analyzing, and visualizing data before building machine learning models.

1. Import Libraries

Why? Libraries help us work with data, mathematics, and visualizations.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set beautiful plot style
sns.set_theme(style="whitegrid")

print("Libraries imported successfully.")
```

Expected Output:

```
Libraries imported successfully.
```

2. Load Dataset

Why? To read the dataset into Python workspace.

Code:

```
df = pd.read_csv("data.csv")

print("Dataset loaded into dataframe.")
```

Expected Output:

```
Dataset loaded into dataframe.
```

3. Understand Dataset

Why? To understand rows, columns, shape, and datatypes.

Code:

```
# First 5 rows
df.head()

# Shape
df.shape

# Information
df.info()
```

Expected Output:

```
Rows and columns displayed successfully.
Data types identified.
Missing values identified.
```

4. Statistical Summary

Why? To understand average, minimum, maximum, and spread of numerical data.

Code:

```
df.describe()
```

Expected Output:

```
Mean → Average value
Min → Smallest value
Max → Largest value
Std → Spread of data
```

5. Missing Value Handling

Why? Missing values reduce model performance and create data gaps.

Code:

```
# Check missing values
df.isnull().sum()

# Fill numerical column
df['Age'] = df['Age'].fillna(df['Age'].mean())

# Fill categorical column
```

```
df['City'] = df['City'].fillna(df['City'].mode()[0])
```

Expected Output:

Missing values handled successfully.

6. Duplicate Values

Why? Duplicate rows create biased analysis.

Code:

```
# Check duplicates
df.duplicated().sum()

# Remove duplicates
df.drop_duplicates(inplace=True)
```

Expected Output:

Duplicate rows removed successfully.

7. Separate Numerical & Categorical Columns

Why? Different data types require different processing techniques.

Code:

```
num_cols = df.select_dtypes(include=['int64', 'float64']).columns

cat_cols = df.select_dtypes(include=['object']).columns

print(num_cols)
print(cat_cols)
```

Expected Output:

Numerical and categorical columns separated.

8. Univariate Analysis

Why? Analyze one column at a time.

Code:

```
# Histogram
sns.histplot(df['Age'], kde=True)

# Boxplot
```

```
sns.boxplot(x=df['Salary'])

plt.show()
```

Expected Output:

Histogram shows distribution.
Boxplot helps detect outliers.

9. IQR Outlier Detection

Why? IQR helps identify abnormal values mathematically.

Code:

```
Q1 = df['Salary'].quantile(0.25)
Q3 = df['Salary'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

print(lower)
print(upper)
```

Expected Output:

Values outside lower and upper limits are outliers.

10. Treat Outliers

Why? Extreme values can negatively affect ML models.

Code:

```
df = df[(df['Salary'] >= lower) &
        (df['Salary'] <= upper)]
```

Expected Output:

Outliers removed successfully.

11. Correlation Heatmap

Why? Understand relationships between numerical features.

Code:

```
corr = df.corr(numeric_only=True)

sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm'
)

plt.show()
```

Expected Output:

+1 → Strong Positive
0 → No Relationship
-1 → Strong Negative

12. Scatter Plot

Why? Visualize relationship between two numerical columns.

Code:

```
sns.scatterplot(
    x='Age',
    y='Salary',
    hue='Gender',
    data=df
)

plt.show()
```

Expected Output:

Relationship between Age and Salary visualized.

13. Encoding

Why? Machine learning models cannot understand text directly.

Code:

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df['Gender'] = le.fit_transform(df['Gender'])
```

Expected Output:

Categorical values converted into numbers.

14. Feature Scaling

Why? Scaling brings all features into the same range.

Code:

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaled_data = scaler.fit_transform(df[num_cols])
```

Expected Output:

Features standardized successfully.

■ Golden Rules in EDA

1. Understand the business problem before coding.
2. Do not remove outliers blindly.
3. Postal Codes and IDs are categorical identifiers.
4. Correlation does not equal causation.
5. EDA is about understanding the hidden story behind the data.

Final Note for Students:

EDA is not just cleaning data. It is the process of understanding the hidden story inside the dataset before building machine learning models.